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Course: Calculus Beta

Assignment: MyLab Math - Uge 12

For the function $f(x,y) = 7x^6 - 2y^3 - 11$, find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

To find the partial derivative for the function $f(x,y) = 7x^6 - 2y^3 - 11$ with respect to x , take the derivative treating y as a constant.

Differentiating the equation $f(x,y)$ with respect to x gives

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(7x^6) - \frac{\partial}{\partial x}(2y^3) - \frac{\partial}{\partial x}(11).$$

Differentiate the first term with respect to x . The partial derivative of $7x^6$ with respect to x is $42x^5$.

Differentiate the second term with respect to x . The partial derivative of $-2y^3$ with respect to x is zero, because y is treated as a constant.

Differentiate the constant term -11 with respect to x . The partial derivative of -11 with respect to x is zero.

The values of the differential terms with respect to x are

$$\frac{\partial}{\partial x}(7x^6) = 42x^5$$

$$\frac{\partial}{\partial x}(-2y^3) = 0$$

$$\frac{\partial}{\partial x}(-11) = 0$$

$$\frac{\partial f}{\partial x} = 42x^5$$

The partial derivative of $f(x,y)$ with respect to x is $\frac{\partial f}{\partial x} = 42x^5$.

Differentiating the equation $f(x,y)$ with respect to y gives

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(7x^6) - \frac{\partial}{\partial y}(2y^3) - \frac{\partial}{\partial y}(11).$$

Differentiate the first term with respect to y . The partial derivative of $7x^6$ with respect to y is zero, because x is treated as a constant.

Differentiate the second term with respect to y . The partial derivative of $-2y^3$ with respect to y is $-6y^2$.

Differentiate the constant term with respect to y . The partial derivative of -11 with respect to y is zero.

The values of the differential terms of $f(x,y)$ with respect to y are

$$\frac{\partial}{\partial y}(7x^6) = 0$$

$$\frac{\partial}{\partial y}(-2y^3) = -6y^2$$

$$\frac{\partial}{\partial y}(-11) = 0$$

$$\frac{\partial f}{\partial y} = -6y^2$$

The partial derivative of $f(x,y)$ with respect to y is $\frac{\partial f}{\partial y} = -6y^2$.